

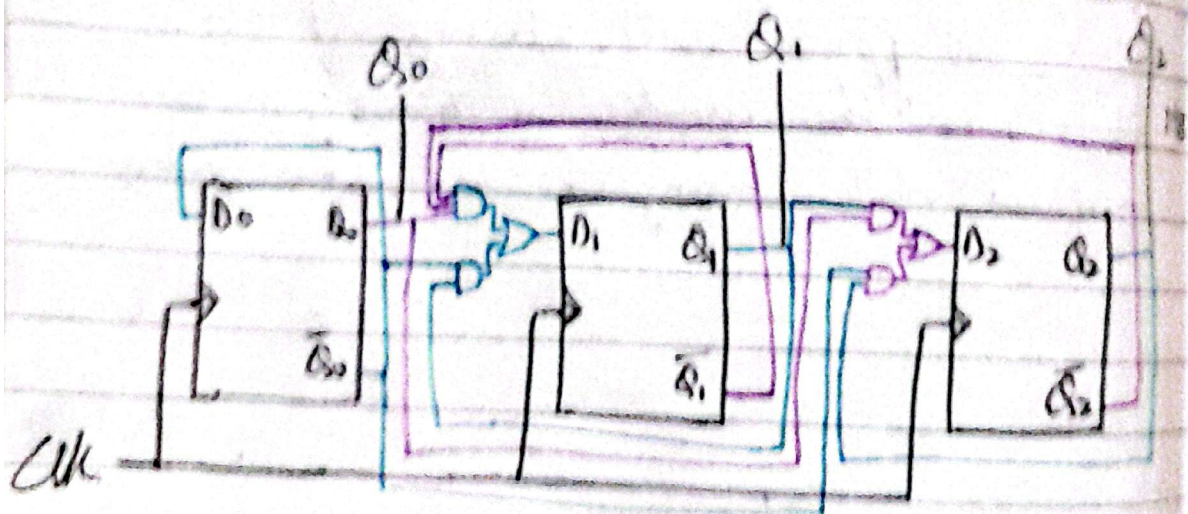
3 a)

Q_2	Q_1	Q_0	Q_2^+	Q_1^+	Q_0^+	D_2	D_1	D_0
0	0	0	0	0	1	0	0	1
0	0	1	0	1	0	0	1	0
0	1	0	0	1	1	0	1	0
0	1	1	1	0	0	1	0	0
1	0	0	1	0	1	1	0	1
1	0	1	0	0	0	0	0	0

6, 7 are don't care

D_2	D_1	D_0
$Q_2 \backslash Q_1 Q_0$ 00 01 11 10 0 1 1 1 1 X X X	$Q_2 \backslash Q_1 Q_0$ 00 01 11 10 0 1 1 1 1 X X X	$Q_2 \backslash Q_1 Q_0$ 00 01 11 10 0 1 1 1 1 1 X X

$$D_2 = Q_1 Q_0 + Q_2 \bar{Q}_0 \quad D_1 = \bar{Q}_2 \bar{Q}_1 Q_0 + Q_1 \bar{Q}_0 \quad D_0 = \bar{Q}_0$$



Potential Glitches

- False transient states
- If the clock period is smaller than the propagation delay, the LSB will toggle again before the MSB changes state

000 $\rightarrow$ 011 $\rightarrow$ 101 $\rightarrow$ 110 $\rightarrow$ 000

Q4 a)

Q_2	Q_1	Q_0	Q_2^+	Q_1^+	Q_0^+	$J_2 K_2$	$J_1 K_1$	$J_0 K_0$
0	0	0	0	1	1	0 X	1 X	1 X
0	1	1	1	0	1	1 X	X 1	X 0
1	0	1	1	1	0	X 0	1 X	X 1
1	1	0	0	0	0	X 1	X 1	0 X

$Q_2 \backslash Q_1 Q_0$		J_2			
		00	01	11	10
0	0	0	X	1	X
1	0	X	X	X	X

$Q_2 \backslash Q_1 Q_0$		K_2			
		00	01	11	10
0	0	X	X	X	X
1	0	X	0	X	1

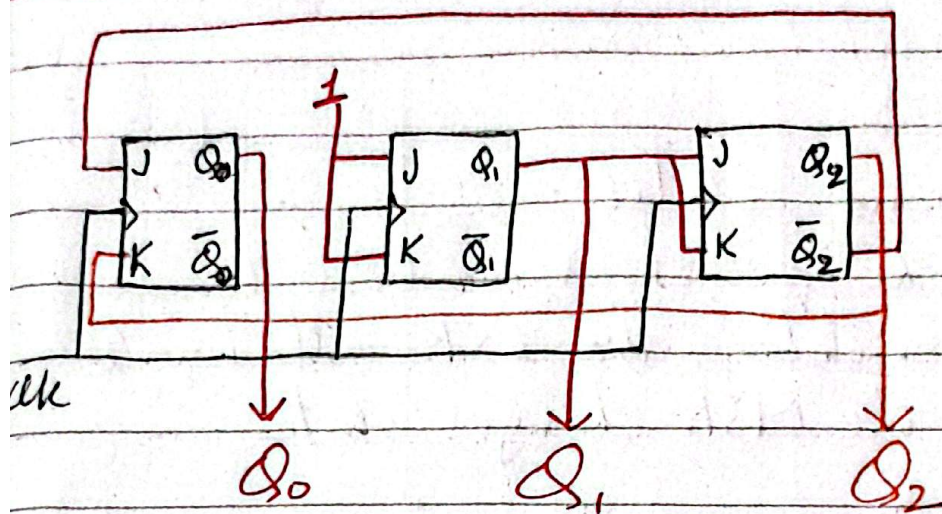
$Q_2 \backslash Q_1 Q_0$		J_1			
		00	01	11	10
0	0	1	X	X	X
1	0	X	1	X	X

$Q_2 \backslash Q_1 Q_0$		K_1			
		00	01	11	10
0	0	X	X	1	X
1	0	X	X	X	1

$Q_2 \backslash Q_1 Q_0$		J_0			
		00	01	11	10
0	0	1	X	X	X
1	0	X	X	X	0

$Q_2 \backslash Q_1 Q_0$		K_0			
		00	01	11	10
0	0	X	X	0	X
1	0	X	1	X	X

$$J_2 = Q_1 \quad K_2 = Q_1 \quad J_1 = 1 \quad K_1 = 1 \quad J_0 = \overline{Q_2} \\ K_0 = Q_2$$



Q_2	Q_1	Q_0	D_2	D_1	D_0	$Q_2^+ = D_2$
0	0	0	0	1	1	$Q_1^+ = D_1$
0	1	1	1	0	1	$Q_0^+ = D_0$
1	0	1	1	1	0	
1	1	0	0	0	0	

Q_2	Q_1	Q_0	
0	0	0	X
0	1	1	X
1	0	1	0
1	1	0	0

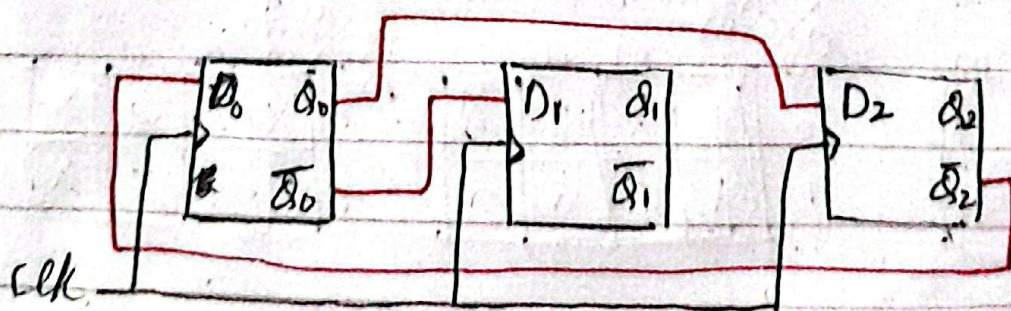
$$D_2 = Q_0$$

Q_2	Q_1	Q_0	
0	0	0	1
0	1	1	X
1	0	1	X
1	1	0	0

$$D_1 = \bar{Q}_1$$

Q_2	Q_1	Q_0	
0	0	0	1
0	1	1	X
1	0	1	X
1	1	0	0

$$D_0 = \bar{Q}_2$$



Q9. b)

M	$\delta_2 \delta_1 \delta_0$	$\delta_2^+ \delta_1^+ \delta_0^+$	$J_2 K_2$	$J_1 K_1$	$J_0 K_0$
0	000	001	0X	0X	1X
0	001	010	0X	1X	X1
0	010	011	0X	X0	1X
0	011	100	1X	X1	X1
0	100	101	X0	0X	1X
0	101	110	X0	1X	X1
0	110	111	X0	X0	1X
0	111	000	X1	X1	X1
1	000	111	1X	1X	1X
1	001	000	0X	0X	X1
1	010	001	0X	X1	1X
1	011	010	0X	X0	X1
1	100	011	X1	1X	1X
1	101	100	X0	0X	X1
1	110	101	X0	X1	1X
1	111	110	X0	X0	X1

$$J_2 = \Sigma (3, 8) + d(4, 5, 6, 7, 12, 13, 14, 15)$$

$$K_2 = \Sigma (7, 12) + d(0, 1, 2, 3, 8, 9, 10, 11)$$

$$J_1 = \Sigma (1, 5, 8, 12) + d(2, 3, 6, 7, 10, 11, 14, 15)$$

$$K_1 = \Sigma (3, 7, 10, 14) + d(0, 1, 4, 5, 8, 12, 9)$$

$$J_0 = \Sigma (0, 2, 4, 6, 8, 10, 12, 14) + d(1, 3, 5, 7, 9, 11, 13, 15)$$

$$K_0 = \Sigma (1, 3, 5, 7, 9, 11, 13, 15) + d(\text{even num})$$

J_2

K_2

$MQ_2 \backslash Q_1 Q_0$	00	01	11	10
00			1	
01	X	X	X	X
11	X	X	X	X
10	1			

$MQ_2 \backslash Q_1 Q_0$	00	01	11	10
00	X	X	X	X
01			1	
11	1			
10	X	X	X	X

$MQ_2 \backslash Q_1 Q_0$	00	01	11	10
00		1	X	X
01		1	X	X
11	1		X	X
10	1		X	X

$MQ_2 \backslash Q_1 Q_0$	00	01	11	10
00	X	X	1	
01	X	X	1	
11	X	X		1
10	X	X		1

$MQ_2 \backslash Q_1 Q_0$	00	01	11	10
00	1	X	X	1
01	1	X	X	1
11	1	X	X	1
10	1	X	X	1

$MQ_2 \backslash Q_1 Q_0$	00	01	11	10
00	X	1	1	X
01	X	1	1	X
11	X	1	1	X
10	X	1	1	X

$$J_2 = M\bar{Q}_B\bar{Q}_C + \bar{M}Q_BQ_C$$

$$K_2 = M\bar{Q}_B\bar{Q}_C + \bar{M}Q_BQ_C$$

$$J_B = \bar{M}Q_C + M\bar{Q}_C$$

$$K_1 = \bar{M}Q_C + M\bar{Q}_C$$

$$J_0 = 1$$

$$K_0 = 1$$

